Sample the posterior: Metropolis-Hasting

Given a set of parameters we can now compute the posterior. How we find the set that maximise it?

5D grid search: not smart

Metropolis-Hasting algorithm!

$q(\theta)$ is the proposal distribution, describes the form of the peak in the parameters space

We chose a 5D multivariate gaussian but how we chose his covariance matrix?

- «Blind» run: we use a diagonal covariance matrix with elements equals to 5% of the bound
- «Not Blind» run: we use a covariance matrix extimated after a previous run

Base idea

- Sample a θ_t from $q(\theta_0)$

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Acceptance rule:

$$\frac{p(\theta_t)}{p(\theta_0)} \cdot \frac{q(\theta_t|\theta_0)}{q(\theta_0|\theta_t)} \geq a$$

Posterior values

Proposal probability ratio
ours is symmetric so it's one

Random number
from [0,1]

Base idea

- Sample a θ_t from $q(\theta_0)$
- Compute the posterior on θ_t
- Check acceptance rule

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- Check acceptance rule

If true move
to $\theta_0 = \theta_t$

If false stay at θ_0

In this way we will have the θ with higher posterior more times so more probable!